RAMCO INSTITUTE OF TECHNOLOGY
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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch: VI Semester B.E.ECE B
Course Code & Title: EC8652 Wireless Communication
Name of the Faculty member (s): Mrs.R.Ramalakshmi

Innovative Practice Description

- **Unit / Topic: Unit III /** OFDM Principle -Implementation of Transceivers, Frequency selective channels, Cyclic Prefix and Windowing
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen:** Flipped Class Room
- **Justification:**

The students will be able to understand the concepts of OFDM Principle - Implementation of Transceivers, Frequency selective channels, Cyclic Prefix and Windowing with the help ofFlipped Class Room activity more easily. The students will be able to easily remember the concepts by watching the videos before teaching the concept.

- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

A flipped classroom is an instructional strategy and a type of blended learning, which aims to increase student engagement and learning by having students complete readings at home and work on live problem-solving during class time.

The flipped classroom is a teaching approach where the students get their first exposure to course content before coming to class through readings and video lectures. They then spend in-class time engaging in activities that have been designed to promote a deeper understanding of a concept. The students learn the concepts of OFDM Principle - Implementation of Transceivers, Frequency selective channels, Cyclic Prefix and Windowing by watching the videos before itself. Then the students were allowed to discuss the topics among their friends. Then students grouped and shared their views on the topic. Since the students discussed among their friends, they understood the concepts more clearly.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO10	PO12	PSO1	PSO3
CO3	3	2	2	3	2	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

	PO1	PO5	PSO1
Innovative practice	3	3	2
Justification for correlation	Flipped Class Room activity improves the learning performance of the students , there will be more interactions within the class room.	The students learn the concepts by watching videos or else through books previously by which there is an enhanced student satisfaction, engagement and enjoyment.	This activity is really useful in learning the concepts of OFDM, Frequency selective channels, Cyclic prefix and windowing by which new technology can be invented.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    The effectiveness of the activity was felt when the students were asked the questions on the topic, they were able to answer easily. They also understood the concepts more easily and able to remember while writing the test.

  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    The students were more interested in learning the concepts if they are sharing the concepts through their friends. They clarified their doubts with the neighboring students and the faculty. They learnt the concepts and asked more doubts in the topic and learnt many things. Outcome attainment has also been increased since the students' performance in the test is also good.
  - ***Challenges faced in implementation:***
    Immediate feedback cannot be got from the students through this activity. There will be a limited student preparation sometimes, more time and work for students.

Textbooks:

1. Rappaport,T.S, -Wireless communications, Pearson Education, Second Edition, 2010.(Unit I,II,IV)
2. Andreas.F. Molisch, -Wireless Communications, John Wiley – India, 2006.(Unit III,V)

Other References:

1. David Tse and Pramod Viswanath - Fundamentals of Wireless Communication, Cambridge University Press, 2005.
2. Wireless Communication- Andrea Goldsmith, Cambridge University Press, 2011
3. Van Nee, R. and Ramji Prasad - OFDM for wireless multimedia communications, Artech House, 2000.
4. Upena Dalal, "Wireless Communication", Oxford University Press, 2009.

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Signature of the Faculty Member

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Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: VI Semester B.E.ECE B
Course Code & Title: EC8652 Wireless Communication
Name of the Faculty member (s): Mrs.R.Ramalakshmi

Innovative Practice Description

- **Unit / Topic: Unit V /** Comparison of beamforming, receiver diversity and transmitter diversity
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 19
- **Activity Chosen:** Reciprocal Peer Questioning
- **Justification:**

  The students will be able to understand the concepts of Intellectual Property Rights with the help of Reciprocal Peer Questioning activity more easily. The students will be able to easily remember the concepts if questions are being asked to them.

- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

Reciprocal Peer Questioning is an activity in which students question each other about the content they are learning using higher-order, open-ended question stems. The questions are used to promote thinking and generate focused discussions in small groups.

The students were in the class room. First the students were taught with the concepts of Comparison of beamforming, receiver diversity and transmitter diversity. Then the students were allowed to discuss the topics among their friends. Then the students were encouraged to ask questions to eachother by which the concepts are learnt easily and more quickly.

- **CO – PO / PSO mapping:**

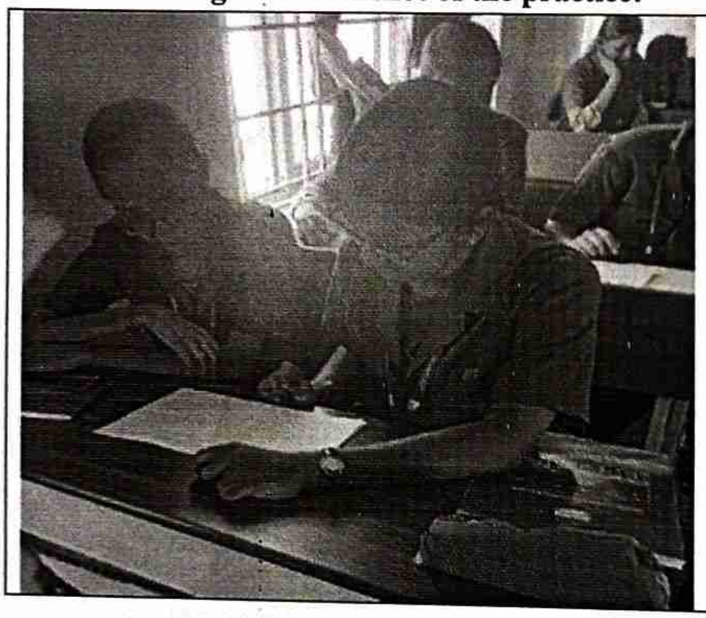
CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO1	PSO3
CO5	3	3	3	2	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PSO1**	**PSO3**
	3	**3**	**3**	**2**	**2**
Justification for correlation	Reciprocal Peer Questioning activity improves the learning performance of the students , there will be more interactions within the class room.	This activity helps the students to analyze the problems in the designing on antennas.	The students get the knowledge on beamforming, diversity concepts by which they will be able to design the antenna.	Reciprocal Peer Questioning activity on beamforming, diversity helps the students to innovate new ideas in the field of communication.	The knowledge of diversity, beamforming concepts will be used in analyzing and designing the antenna and its related products.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    The effectiveness of the activity was felt when the students were asked the questions on the topic, they were able to answer easily. They also understood the concepts more easily and able to remember while writing the test.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    The students were more interested in learning the concepts if they are asking questions and sharing the concepts through their friends. They clarified their doubts with the neighboring students and the faculty. They learnt the concepts and asked more doubts in the topic and learnt many things. Outcome attainment has also been increased since the students' performance in the test is also good.

- ***Challenges faced in implementation:***

  Basically some time is needed to explain the basic concepts of Beamforming, Diversity to the students and also to discuss among the students.

Textbooks:

3. Rappaport,T.S, -Wireless communications, Pearson Education, Second Edition, 2010.(Unit I,II,IV)
4. Andreas.F. Molisch, -Wireless Communications, John Wiley – India, 2006.(Unit III,V)

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Active Learning Activity

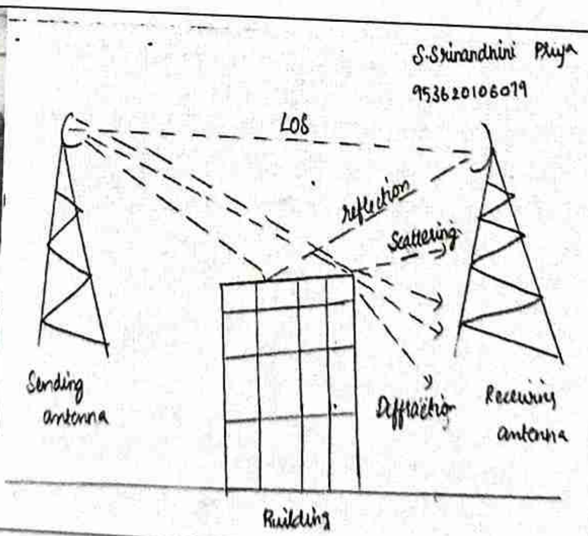
Degree, Semester & Branch: VI Semester B.E.ECE B
Course Code & Title: EC8652 Wireless Communication
Name of the Faculty member (s): Mrs.R.Ramalakshmi

Unit II - Cellular Architecture

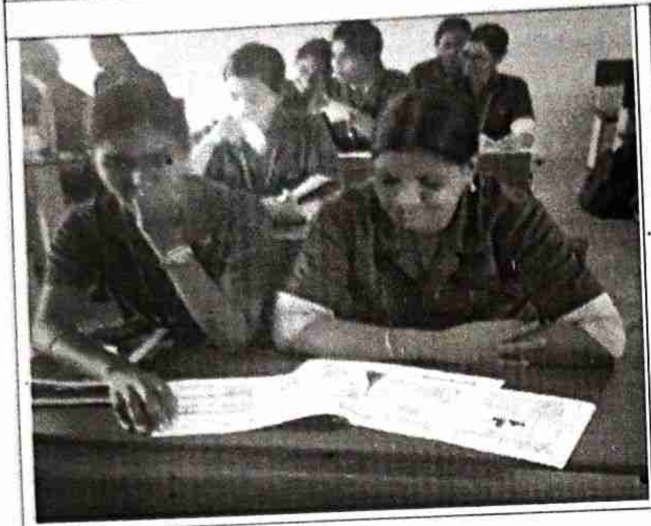
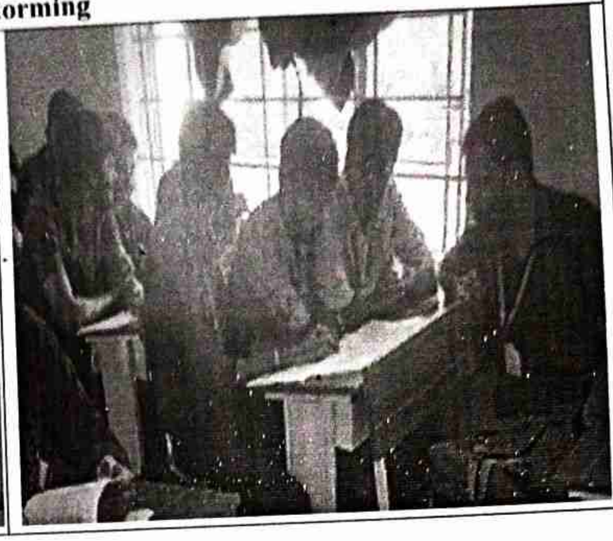
Think Pair Share Activity

Unit I - Wireless Channels

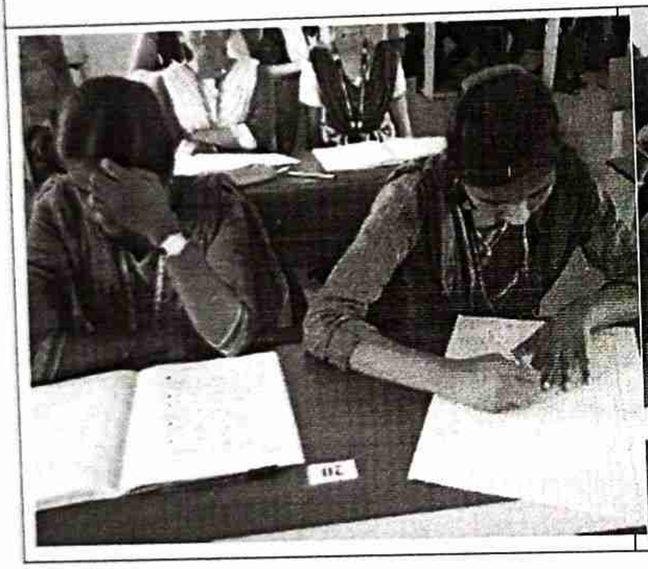
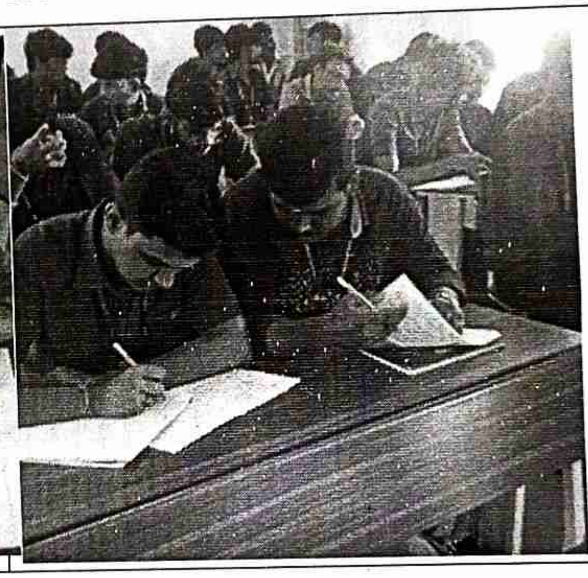
Sketch Noting Activity



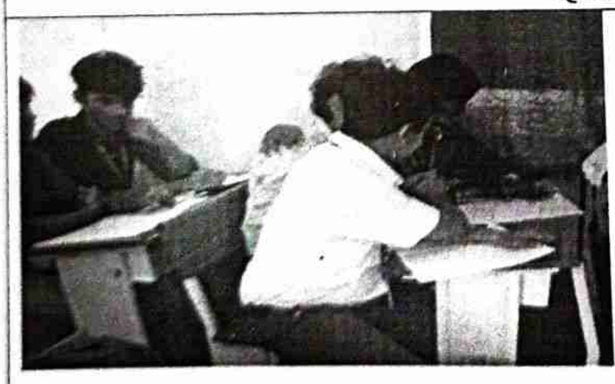
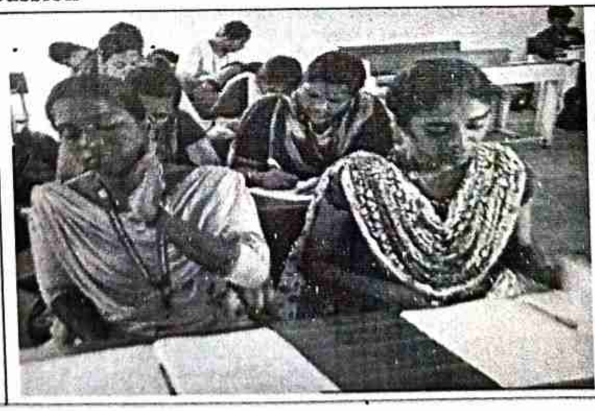
Unit III – Digital Signaling for Fading Channels
Brainstorming

Unit IV - Multipath Mitigation Techniques
Open Book Test

Unit V – Multiple Antenna Techniques
Quescussion

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